Getting Started with App

Hectron

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Contents

1 Description

Document informs the user how to start the application residing in the nba_app directory of repository ReactApps. Applications were developed on MacOS *Sonoma*. The following programming languages were used throughout fullstack development.

- **FrontEnd:**
  Typescript - imperative
- **Backend**:
  Postgres - declarative
- **Script**:
  Bash - imperative

2 Run Application

- Open Docker. If Docker is not installed, please download at `https://www.docker.com/products/docker-desktop/`
- Start/Configure Postgres Database (Docker Container Instance)
  - Change directory to nba_app/database
  - run run.sh
- Start Monitor Service.
  The highest priority service. All request are handled by the monitor service. The start monitor must be running for proper operation of the application.
  - Change directory to nba_app
  - run server/main.sh *Monitor service is now running*
- Open web browser; request application at address and port `http://127.0.0.1:50214/`

3 Figures

Figure 1: Home Page

Figure 2: **GET** NBA Service

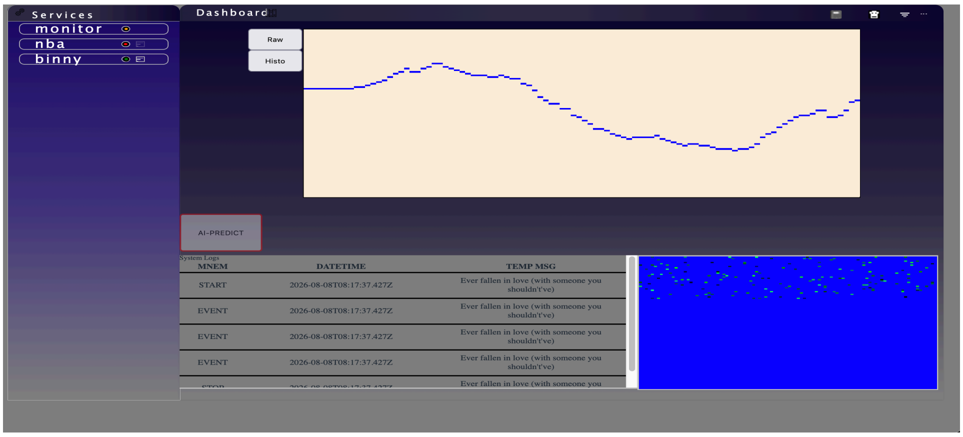

Figure 3: **GET** Binny Service - XY Plot

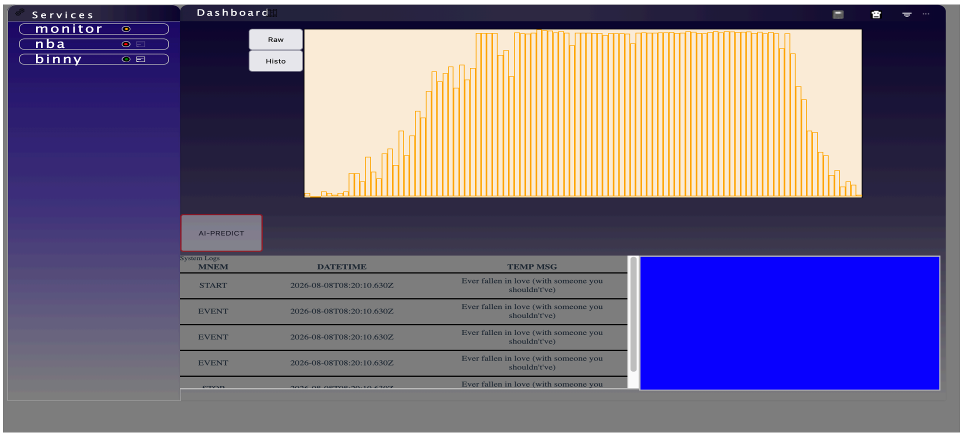

Figure 4: **GET** Binny Service - Histo

4 Load Temperature Train Data into DB

We used temperature data from Kaggle https://www.kaggle.com/datasets/bhanupratapbiswas/weather-data to represent log data. The prediction model will be used predict potential future logs; this will be discussed in a future section.

4.1 Loading data into database

- Download WeatherData.csv aforementioned above and lowercase the filename.
- Move weatherdata.csv into nba_app\database\pgfiles directory
- Create sql file (e.g. index2.sql) with schema matching the features labeled in WeatherData.csv

```
CREATE TABLE  binny.temp (
    date_time text,
    temp real ,
    dew_point_temp real,
    humidity decimal,
    wind_speed decimal,
    visibility decimal,
    weather text
    );
```

- Create script copyweather.sh in nba_app\database\pgfiles
  copyweather.sh

```
DB=sport_db
USER=htron
psql  -d "${DB}" --username "${USER}"  <<< "\copy binny.temp \
FROM '/root/work/WeatherData.csv' DELIMITER ',' CSV HEADER"
```

- Add the following commands to nba_app\database\run.sh
  run.sh

```
DB=sport_db
USER=htron
docker exec "${CONTAINER_ID}" psql -d "${DB}"  --username "${USER}" -f /root/work/index2.sql

docker cp ./pgfiles/copyweather.sh \
<CONTAINER_ID\>:/root/work/copyweather.sh

docker cp ./pgfiles/weatherdata.csv \
<CONTAINER_ID\>:/root/work/weatherdata.csv

stdout=$(docker exec "${CONTAINER_ID}" \
./root/work/copyweather.sh & )
echo $stdout
```

5 View Database Table

Let's view the weather table

- Access PostgreSQL container (i.e. username=htron)

```
docker exec -it nba-pg psql -d sport\db --username htron
```

- Verify binny table exist

```
sport\_db=> \dt binny.*

    List of tables
Schema | Name | Type  | Owner
--------+------+-------+-------
binny  | temp | table | htron


sport_db=> select * from binny.temp;

    date_time     | temp  | dew_point_temp | humidity | wind_speed | visibility |  kpa   |                 weather
------------------+-------+----------------+----------+------------+------------+--------+-----------------------------------------
 1/1/2012 0:00    |  -1.8 |           -3.9 |       86 |          4 |          8 | 101.24 | Fog
 1/1/2012 1:00    |  -1.8 |           -3.7 |       87 |          4 |          8 | 101.24 | Fog
 1/1/2012 2:00    |  -1.8 |           -3.4 |       89 |          7 |          4 | 101.26 | Freezing Drizzle,Fog
 1/1/2012 3:00    |  -1.5 |           -3.2 |       88 |          6 |          4 | 101.27 | Freezing Drizzle,Fog
 1/1/2012 4:00    |  -1.5 |           -3.3 |       88 |          7 |        4.8 | 101.23 | Fog
 1/1/2012 5:00    |  -1.4 |           -3.3 |       87 |          9 |        6.4 | 101.27 | Fog
 1/1/2012 6:00    |  -1.5 |           -3.1 |       89 |          7 |        6.4 | 101.29 | Fog
 1/1/2012 7:00    |  -1.4 |           -3.6 |       85 |          7 |          8 | 101.26 | Fog
 1/1/2012 8:00    |  -1.4 |           -3.6 |       85 |          9 |          8 | 101.23 | Fog
 1/1/2012 9:00    |  -1.3 |           -3.1 |       88 |         15 |          4 |  101.2 | Fog
 1/1/2012 10:00   |    -1 |           -2.3 |       91 |          9 |        1.2 | 101.15 | Fog
 1/1/2012 11:00   |  -0.5 |           -2.1 |       89 |          7 |          4 | 100.98 | Fog
 1/1/2012 12:00   |  -0.2 |             -2 |       88 |          9 |        4.8 | 100.79 | Fog
 1/1/2012 13:00   |   0.2 |           -1.7 |       87 |         13 |        4.8 | 100.58 | Fog
 1/1/2012 14:00   |   0.8 |           -1.1 |       87 |         20 |        4.8 | 100.31 | Fog
 1/1/2012 15:00   |   1.8 |           -0.4 |       85 |         22 |        6.4 | 100.07 | Fog
 1/1/2012 16:00   |   2.6 |           -0.2 |       82 |         13 |       12.9 |  99.93 | Mostly Cloudy
 1/1/2012 17:00   |     3 |              0 |       81 |         13 |       16.1 |  99.81 | Cloudy
 1/1/2012 18:00   |   3.8 |              1 |       82 |         15 |       12.9 |  99.74 | Rain
 1/1/2012 19:00   |   3.1 |            1.3 |       88 |         15 |       12.9 |  99.68 | Rain
 1/1/2012 20:00   |   3.2 |            1.3 |       87 |         19 |         25 |   99.5 | Cloudy
```